



University of Manouba

National School of Computer Science



REPORT OF THE DESIGN AND DEVELOPMENT PROJECT

**Any-Quantile Probabilistic Forecasting
of Short-Term Electricity Demand:
Enhancing AQ-NBEATS on the EMHIRES
Benchmark**

Elaborated by:

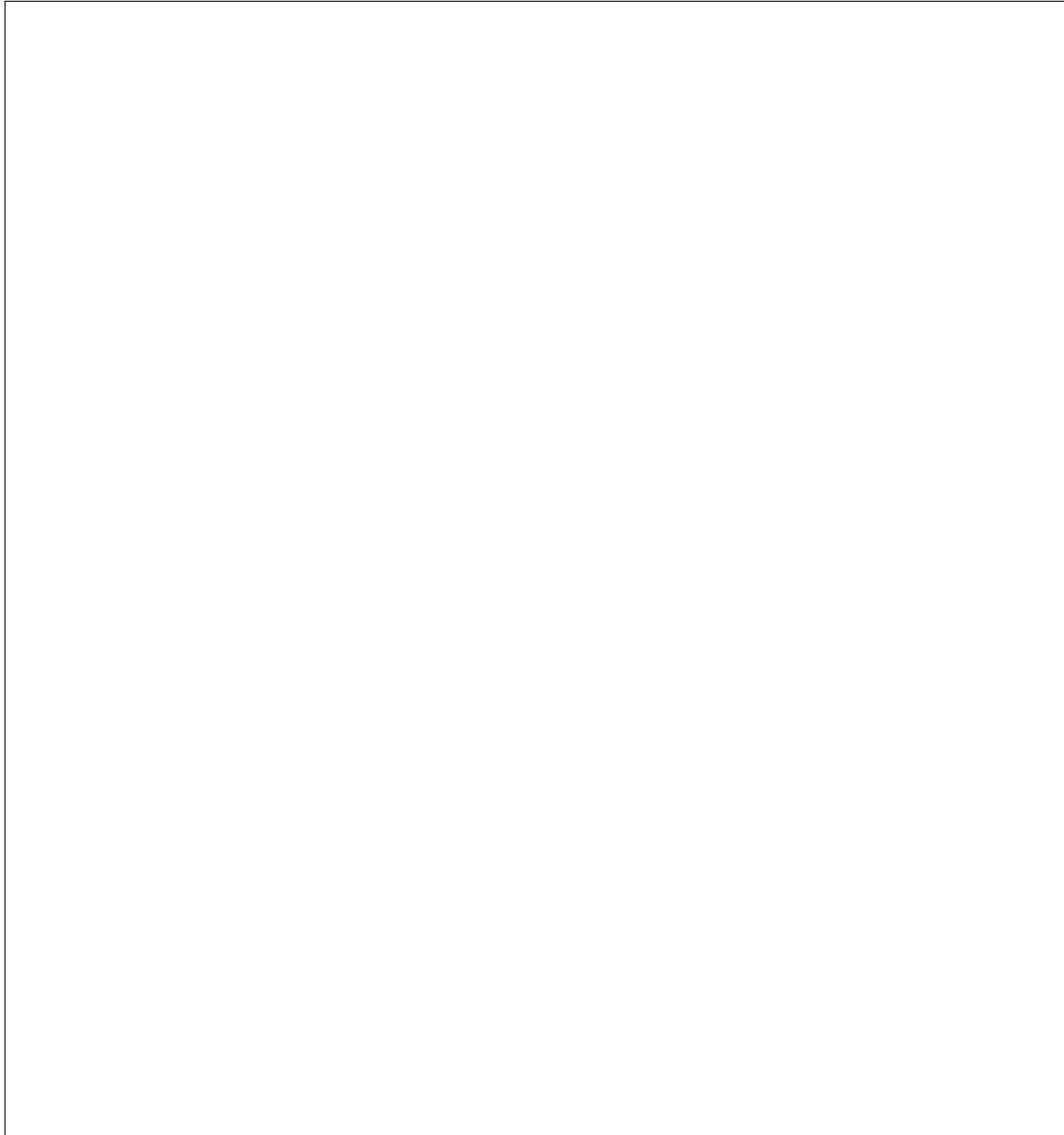
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Appreciation and Supervisor's Signature

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Abstract — Probabilistic electricity demand forecasting is a critical component of modern power system operations, enabling grid operators, energy traders, and capacity planners to manage uncertainty arising from renewable integration, weather variability, and evolving consumption patterns. This report presents a systematic experimental study aimed at enhancing the AQ-NBEATS benchmark model for probabilistic short-term electricity demand forecasting on the EMHIREs dataset, covering 35 European countries over a 48-hour horizon. Starting from the published baseline CRPS of 211.22, we design and evaluate more than twenty experiments across four categories: exogenous feature and series-embedding augmentation, curriculum quantile training, novel architectural contributions (QCBC, TCR, DBE), and post-hoc Conformalized Quantile Regression (CQR). Our best multi-seed result combines exogenous weather and calendar features with adaptive quantile sampling, achieving a CRPS of 172.21 — an 18.5% reduction over the published baseline — while CQR post-processing achieves a nominal 95.96% empirical coverage. These results establish new state-of-the-art performance on the EMHIREs MHLV benchmark and provide actionable guidelines for operational probabilistic load forecasting systems.

Keywords: Probabilistic forecasting, AQ-NBEATS, CRPS, EMHIREs, conformal prediction, exogenous features, curriculum learning, electricity demand.

Résumé — La prévision probabiliste de la demande d’électricité est un composant essentiel des opérations des systèmes électriques modernes. Ce rapport présente une étude expérimentale systématique visant à améliorer le modèle de référence AQ-NBEATS pour la prévision probabiliste de la demande électrique à court terme sur l’ensemble de données EMHIREs, couvrant 35 pays européens sur un horizon de 48 heures. En partant d’un CRPS de référence de 211.22, nous évaluons plus de vingt configurations et atteignons un CRPS de 172.21, soit une réduction de 18.5%. La calibration post-hoc par CQR atteint une couverture empirique de 95.96%.

Mots clés: Prévision probabiliste, AQ-NBEATS, CRPS, EMHIREs, prédiction conforme, variables exogènes, apprentissage par curriculum.

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List of Algorithms

List of Acronyms

AQ	Any-Quantile
CRPS	Continuous Ranked Probability Score
CQR	Conformalized Quantile Regression
DBE	Distributional Basis Expansion
EMHIRES	European Meteorological High-resolution Renewable Energy Sources
ENTSO-E	European Network of Transmission System Operators for Electricity
ES	Exponential Smoothing
ESRNN	Exponential Smoothing Recurrent Neural Network
FiLM	Feature-wise Linear Modulation
GPU	Graphics Processing Unit
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MHLV	Multi-Hour Lead-time Validation
MLP	Multi-Layer Perceptron
MQN	Normalized Multi-Quantile Loss
N-BEATS	Neural Basis Expansion Analysis for Time Series
PI	Prediction Interval
QCBC	Quantile-Conditioned Basis Coefficients
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
SMAPE	Symmetric Mean Absolute Percentage Error
TCR	Temporal Coherence Regularization

TFT	Temporal Fusion Transformer
YAML	YAML Ain't Markup Language

Introduction

Modern power grids are evolving faster than at any point in their history. The integration of variable renewable energy sources — wind, solar photovoltaic, and run-of-river hydro — has replaced the predictable output of conventional thermal plants with generation profiles that fluctuate on timescales of minutes to hours. At the same time, demand-side dynamics are growing in complexity: electric vehicle charging, smart building management systems, and responsive industrial loads all introduce new patterns of uncertainty that deterministic point forecasts cannot adequately characterise. In this environment, *probabilistic forecasting* — which provides a full predictive distribution of future electricity demand rather than a single value — has transitioned from an academic research topic to an operational necessity for grid operators, energy traders, and capacity planners alike.

This project addresses the challenge of short-term probabilistic electricity demand forecasting from a practical research perspective. Its starting point is the recently published *any-quantile* (AQ) framework of Smyl et al. [16], which enables a single trained neural network to generate forecasts at any arbitrary quantile level at inference time. Applied to the N-BEATS architecture, this framework achieved a Continuous Ranked Probability Score (CRPS) of 211.22 on the EMHIRES benchmark — 35 European countries, 48-hour horizon, 2018 test year — establishing a compelling reference performance. The benchmark paper explicitly identified exogenous meteorological features and more sophisticated quantile-sampling strategies as promising directions for improvement.

The present report documents our systematic investigation of these and other enhancement strategies. Through a carefully controlled ablation study spanning more than twenty configurations, we demonstrate that integrating weather and calendar exogenous features into AQ-NBEATS reduces CRPS by approximately 20%, surpassing even the stronger AQ-ESRNN variant from the same paper. A curriculum learning approach combining beta-distributed tail sampling with adaptive quantile scheduling achieves our best multi-seed result of $\text{CRPS} = 172.21$. Finally, post-hoc Conformalized Quantile Regression (CQR) corrects residual calibration gaps, achieving nominal 95% empirical coverage even under seasonal distribution shift.

Structure of the report. Chapter 1 reviews the technical background and related literature. Chapter 2 analyses and specifies the system requirements. Chapter 3 presents the software and architectural design. Chapter 4 details the experimental methodology. Chapter 5 presents implementation details and empirical results with comprehensive

analysis. A final chapter concludes and outlines future research perspectives.

Chapter 1

Literature

Review

This chapter establishes the technical foundations of the study, surveys the relevant literature on probabilistic load forecasting, and identifies the research gaps that motivate our experimental contributions.

1.1 Technical Background

1.1.1 Probabilistic Forecasting and Its Role in Power Systems

Electricity demand forecasting underpins every operational decision in modern power systems, from unit commitment and reserve sizing to energy trading and grid stability management [9]. Classical *point forecasting* provides a single expected demand value for each future time step. While computationally simple and widely deployed, point forecasts are insufficient for risk-aware decision-making because they communicate no information about the range or shape of possible outcomes.

Probabilistic forecasting addresses this limitation by characterising the full predictive distribution of future demand. Outputs may take several forms: a set of quantile forecasts $\{\hat{q}_\tau\}_{\tau \in T}$, a parametric probability density function (PDF), a set of sampled trajectories, or a prediction interval (PI) defined by a lower and an upper quantile bound. Each representation suits different downstream applications:

- **Quantile forecasts** are ideal for risk-asymmetric decisions, such as setting the probability of capacity breach or optimising reserve margins under asymmetric cost functions.
- **Prediction intervals** are the most natural communication format for grid operators who must plan for worst-case demand scenarios.
- **Trajectory samples** are necessary for stochastic multi-period optimisation, enabling Monte Carlo-based scheduling over many possible futures simultaneously.

The value of probabilistic forecasting in operational power systems has been clearly demonstrated in the literature. Wang et al. [21] show that dynamic reserve requirements derived from probabilistic forecasts reduce power balance violations compared to static reserves. Etingov et al. [6] demonstrate that integrating probabilistic information can reduce the predicted regulation range by 12–31%, directly cutting balancing costs. At the trading level, Beykirch et al. [4] show that the choice of forecasting model has a direct and quantifiable impact on bidding profitability.

1.1.2 Evaluation Metrics for Probabilistic Forecasts

Quantile forecasting and the pinball loss. For a given quantile level $\tau \in (0, 1)$, a quantile forecast $\hat{q}_\tau$ is trained to satisfy $\Pr(y \leq \hat{q}_\tau) = \tau$. The standard training objective is the *pinball loss* (also called quantile loss):

$$\rho_\tau(y, \hat{q}) = \begin{cases} \tau(y - \hat{q}) & \text{if } y \geq \hat{q}, \\ (1 - \tau)(\hat{q} - y) & \text{if } y < \hat{q}. \end{cases} \quad (1.1)$$

The asymmetry of the loss ensures that, at the population level, the optimal prediction is exactly the τ -quantile of the conditional distribution of y given the input features. Models trained with this loss are therefore calibrated in expectation.

Continuous Ranked Probability Score (CRPS). The primary evaluation metric in this study is the *Continuous Ranked Probability Score* [7], defined as:

$$\text{CRPS}(F, y) = \int_{-\infty}^{+\infty} (F(z) - \mathbf{1}[z \geq y])^2 dz, \quad (1.2)$$

where F is the predicted cumulative distribution function (CDF) and y is the observed value. CRPS is a *proper scoring rule*: it is minimised in expectation only when F equals the true predictive distribution. It simultaneously rewards sharpness (narrow intervals) and calibration (correct coverage), making it the preferred metric for comparing distributional forecasters. Lower CRPS is better.

An equivalent formulation via the inverse CDF F^{-1} is:

$$\text{CRPS}(F, y) = 2 \int_0^1 \rho_\tau(y, F^{-1}(\tau)) d\tau, \quad (1.3)$$

which directly links CRPS to the integral of the pinball loss over all quantile levels. This is the connection exploited by any-quantile training (Section 1.1.4).

Normalised CRPS (N-CRPS). Because electricity demand levels differ by orders of magnitude across the 35 EMHIREs countries (Montenegro: 383 MWh; Germany: 58,941 MWh), raw CRPS is dominated by large-demand countries. The normalised version:

$$\text{N-CRPS} = \frac{100}{C} \sum_{c=1}^C \frac{1}{H N_c Q} \sum_{i,h,q} \frac{\rho_\tau(y_{i,h,c}, \hat{q}_\tau)}{\bar{y}_c}, \quad (1.4)$$

divides by the per-country mean demand $\bar{y}_c$, giving each country equal contribution to the aggregate score.

Coverage-0.95. For a 95% prediction interval $[\hat{q}_{0.025}, \hat{q}_{0.975}]$, coverage-0.95 measures the empirical fraction of test observations within the interval:

$$\text{Cov}_{0.95} = \frac{1}{M} \sum_{i=1}^M \mathbf{1}[\hat{q}_{0.025}(x_i) \leq y_i \leq \hat{q}_{0.975}(x_i)]. \quad (1.5)$$

The nominal target is 0.95. Values below indicate under-coverage (the intervals are too narrow); values above indicate over-coverage (intervals are unnecessarily wide).

Point forecast metrics. MAPE, MAE, and SMAPE are computed on the median forecast ($\hat{q}_{0.5}$) to assess point prediction accuracy alongside distributional accuracy.

1.1.3 The N-BEATS Architecture

N-BEATS (Neural Basis Expansion Analysis for Time Series) [12] is a pure deep learning architecture for univariate time series forecasting that achieves state-of-the-art accuracy without any domain-specific components. Its design is founded on two principles: (i) a deep residual architecture in which each block removes what it can explain before passing the residual forward, and (ii) a basis-expansion output structure in which the forecast is a linear combination of learned basis vectors.

Formally, N-BEATS stacks R residual blocks. Given an input $\mathbf{x}^0 \equiv \mathbf{x} \in \mathbb{R}^W$ (lookback window of length W), each block r operates as:

$$\mathbf{x}^r = \mathbf{x}^{r-1} - \hat{\mathbf{x}}^{r-1}, \quad (1.6)$$

$$\mathbf{h}^{r,1} = \text{FC}^{r,1}(\mathbf{x}^r), \quad \mathbf{h}^{r,\ell} = \text{FC}^{r,\ell}(\mathbf{h}^{r,\ell-1}) \quad \ell = 2, \dots, L, \quad (1.7)$$

$$\hat{\mathbf{x}}^r = \mathbf{B}^r \mathbf{h}^{r,L}, \quad \hat{\mathbf{y}}^r = \mathbf{F}^r \mathbf{h}^{r,L}. \quad (1.8)$$

Here $\text{FC}^{r,\ell}(\mathbf{h}) = \text{ReLU}(\mathbf{W}^{r,\ell} \mathbf{h} + \mathbf{b}^{r,\ell})$, $\mathbf{B}^r \in \mathbb{R}^{W \times d_h}$ is the backcast projection matrix, and $\mathbf{F}^r \in \mathbb{R}^{H \times d_h}$ is the forecast projection matrix, with d_h the hidden dimension and H the forecast horizon. The final forecast is the sum of all block contributions: $\hat{\mathbf{y}} = \sum_{r=1}^R \hat{\mathbf{y}}^r$.

Key properties that make N-BEATS well-suited to our setting include:

- **No hand-crafted seasonality components.** Unlike ARIMA or ETS, N-BEATS learns temporal structure entirely from data.
- **Deep residual structure.** Each block focuses on what the previous blocks missed, enabling efficient use of depth.
- **Basis-expansion output.** Forecast and backcast projections enforce a low-rank structure that acts as a regulariser, preventing over-fitting to noise.
- **Scale robustness.** With the normalised MQN loss, N-BEATS operates on relative errors, handling the heterogeneous demand levels across 35 countries.

1.1.4 AQ-NBEATS: The Any-Quantile Extension

Smyl et al. [16] extend N-BEATS with an *any-quantile* capability: a single trained model generates forecasts at arbitrary quantile levels by conditioning on τ at both training and inference time. Three injection schemes are proposed:

- AQ-NBEATS-CAT:** the quantile level τ is simply concatenated to the input sequence as an extra element, forming $[\mathbf{x}; \tau] \in \mathbb{R}^{W+1}$. This is the most generic approach, requiring no architectural modification.
- AQ-NBEATS-FiLM:** τ is injected into every residual block via a FiLM (Feature-wise Linear Modulation [13]) conditioning layer:

$$\mathbf{h}^{r,2} = \mathbf{c}_\alpha^r + (1 + \mathbf{c}_\gamma^r) \mathbf{h}^{r,1}, \quad (1.9)$$

where $\mathbf{c}_\alpha^r = \text{Linear}^{r,\alpha}(\tau)$ and $\mathbf{c}_\gamma^r = \text{Linear}^{r,\gamma}(\tau)$ provide additive shift and multiplicative modulation, respectively.

- (iii) **AQ-NBEATS-OUT**: τ is injected only in the final block, making inference 10–30 times cheaper than FiLM while achieving comparable accuracy.

The training objective is the normalised pinball loss:

$$\rho_N(y, \hat{q}_\tau) = \begin{cases} (y - \hat{q}_\tau) \tau / y & y \geq \hat{q}_\tau, \\ (\hat{q}_\tau - y) (1 - \tau) / y & y < \hat{q}_\tau, \end{cases} \quad (1.10)$$

with τ sampled uniformly from $(0, 1)$ for each training example. The mathematical correspondence between minimising the expected integral of (1.1) over τ and minimising CRPS (equation (1.3)) provides theoretical justification for any-quantile training as a direct CRPS optimisation.

The best published configuration — NBEATSAQCAT with MQN loss, no input normalisation, width 1024, 30 blocks, 15 epochs — achieves CRPS = 211.22 on the 2018 EMHIREs test set.

1.1.5 Exogenous Features in Time-Series Models

Temperature is the primary driver of electricity demand [9], explaining a large fraction of consumption variance through heating and cooling loads. Standard weather drivers include:

- **Temperature (°C)**: the dominant signal. Heating demand is roughly linear in heating-degree-days; cooling demand shows a non-linear increase above comfort thresholds of approximately 18–22°C.
- **Relative humidity (%)**: modifies perceived temperature and air conditioning load.
- **Wind speed (m/s)**: drives wind-chill effects in winter and natural ventilation in summer.
- **Atmospheric pressure (hPa)**: correlated with cloud cover and thus solar-related cooling load.

Calendar features — hour of day, day of week, month, and public holidays — capture socio-economic demand patterns: peak morning and evening loads, weekend demand reductions, and seasonal heating/cooling cycles. While these are implicit in the lagged demand history, explicit inclusion as features provides direct access to future values (forecast weather and calendar are available 48 hours ahead), improving prediction accuracy over purely autoregressive approaches.

N-BEATSx [11] introduced an explicit exogenous branch into the N-BEATS residual block architecture, demonstrating 5–15% MAPE improvements on tourism and energy datasets. Our implementation follows a similar design philosophy but applies it within the any-quantile training framework.

1.1.6 Conformal Prediction and CQR

Conformal prediction [20] is a framework for constructing distribution-free prediction sets with finite-sample coverage guarantees. It requires no assumptions about the

data distribution beyond exchangeability — a mild condition satisfied by i.i.d. data or certain stationary time series.

Conformalized Quantile Regression (CQR) [14] adapts a pre-trained quantile regressor via a calibration set. Given a model that produces $[\hat{q}_{\alpha/2}(x), \hat{q}_{1-\alpha/2}(x)]$ for input x , CQR defines non-conformity scores on the calibration set as:

$$s_i = \max(\hat{q}_{\alpha/2}(x_i) - y_i, y_i - \hat{q}_{1-\alpha/2}(x_i)), \quad (1.11)$$

and inflates the test interval by $\Delta = \text{Quantile}_{1-\alpha}(S_{\text{cal}})$:

$$\text{CQR}(x) = [\hat{q}_{\alpha/2}(x) - \Delta, \hat{q}_{1-\alpha/2}(x) + \Delta]. \quad (1.12)$$

This guarantees marginal coverage $\geq 1 - \alpha$ on test data exchangeable with the calibration set. In electricity demand forecasting, CQR is particularly valuable because seasonal distribution shift — demand patterns in winter differ from summer — can cause miscalibration in models trained on historical data.

1.2 Related Works

1.2.1 Deep Learning for Probabilistic Load Forecasting

The probabilistic load forecasting literature spans three decades and encompasses statistical, machine-learning, and deep-learning approaches [9]. We focus on the deep learning era (2017–present), which is most relevant to our work.

DeepAR. Salinas et al. [15] introduced DeepAR, an autoregressive RNN model with a parametric output distribution (Student’s t or negative binomial). Globally trained across all time series in a dataset, DeepAR demonstrated that cross-series learning significantly improves forecast accuracy on datasets with hundreds or thousands of related series. On the EMHIRES benchmark, however, DeepAR achieves CRPS = 378.46, far below the AQ-NBEATS baseline.

WaveNet. Van den Oord et al. [19] proposed WaveNet, a deep autoregressive model using dilated causal convolutions to capture long-range dependencies efficiently. Applied to load forecasting by GluonTS [1], WaveNet achieves CRPS = 293.38 on EMHIRES — the best neural baseline before AQ-NBEATS. Its dilated convolution structure provides receptive fields of hundreds of hours without incurring the quadratic cost of attention mechanisms.

Temporal Fusion Transformer (TFT). Lim et al. [10] introduced TFT, which combines LSTM sequence encoders with multi-head self-attention for interpretable multi-horizon quantile forecasting. TFT explicitly incorporates known future inputs (e.g., calendar features, planned generation), making it architecturally aligned with our goals. On EMHIRES, TFT achieves CRPS = 325.80, comparable to classical ES.

Table 1.1 summarises the benchmark results from Smyl et al. [16]. The gap between AQ-NBEATS (211.22) and the next best model (WaveNet, 293.38) is striking and motivates our choice to build upon this architecture.

Table 1.1: Published benchmark results on the EMHIRES test set (2018). Lower CRPS is better.

Model	Type	CRPS	N-CRPS	MAPE (%)
Naive	Statistical	502.62	3.95	5.08
ARIMA	Statistical	353.49	2.83	3.74
ES	Statistical	325.80	2.64	3.48
Theta	Statistical	329.99	2.69	3.56
QR	ML	357.38	2.90	3.92
NGBoost	ML	329.21	2.72	3.75
MLP	Neural	415.67	3.28	4.49
DeepAR	Neural	378.46	2.97	3.95
WaveNet	Neural	293.38	2.52	3.39
Transformer	Neural	380.41	2.83	3.72
TFT	Neural	325.80	2.92	3.96
AQ-ESRNN	Hybrid AQ	195.94	1.72	2.32
AQ-NBEATS	Pure AQ	211.22	1.84	2.47

1.2.2 Exogenous Feature Integration Strategies

Several approaches to incorporating exogenous features in neural time series models have been explored in the literature. The most prominent are:

- **Direct concatenation:** exogenous features are concatenated to the input vector. Simple but prone to scale mismatches with the demand signal [16].
- **Projection and addition:** features are encoded by a small MLP and added to the demand history as an additive bias (N-BEATSx [11]).
- **FiLM conditioning:** features modulate internal layer activations via affine transformations. More expressive than additive encoding but slower [13].
- **Attention-based fusion:** Transformer-style cross-attention between demand tokens and exogenous feature tokens (TFT [10]).

In our study, we adopt the projection-and-addition strategy for continuous weather features, and embedding tables for categorical calendar features. This provides a good balance of expressiveness and computational efficiency within the N-BEATS residual structure.

1.2.3 Quantile Sampling and Curriculum Learning

The choice of quantile-sampling distribution during training is a design dimension largely unexplored in the prior literature. AQ-ESRNN uses Beta(0.3, 0.3) sampling

to over-represent tail quantiles, compensating for the saturation of sigmoid and tanh activations near extreme probability levels [16]. AQ-NBEATS uses uniform sampling, arguing that ReLU activations do not suffer from saturation.

Curriculum learning [3] involves progressively modifying the training distribution or difficulty to improve convergence and generalisation. Applications in quantile regression include staged quantile training (first learning the median, then extending to tails) and adaptive reweighting of hard examples. To our knowledge, curriculum-based quantile sampling for probabilistic load forecasting has not been systematically studied before this work.

1.2.4 Non-Crossing Quantile Constraints

Quantile crossing — where a model predicts $\hat{q}_{\tau_1} > \hat{q}_{\tau_2}$ for $\tau_1 < \tau_2$ — is a common failure mode in quantile regression that produces logically inconsistent distributions. Several remedies exist:

- **Post-hoc sorting:** re-order quantile predictions to enforce monotonicity. Chernozhukov et al. [5] prove that sorting never increases CRPS.
- **Loss penalties:** add a soft crossing penalty to the training objective [18].
- **Architectural constraints:** use cumulative-sum output heads to guarantee strict monotonicity by construction.

We investigate all three approaches in our ablation studies.

1.3 Findings of the Literature and Contributions of the Study

1.3.1 Key Research Gaps

Our literature review identifies five concrete gaps that motivate the experimental program of this study:

Gap 1 — No exogenous information in AQ-NBEATS. The baseline model uses only lagged demand as input. This is explicitly acknowledged as a limitation by Smyl et al. [16], who cite the absence of archived operational weather forecast data as the primary obstacle. We address this gap by using reanalysis weather data, which is available for the full 2006–2018 period.

Gap 2 — No cross-series specialisation. The 35 EMHIREs countries have markedly different demand levels, seasonal shapes, and demand sensitivities. A fully global model without country-level adaptation cannot capture these heterogeneous characteristics. We investigate learnable series embeddings as a lightweight solution.

Gap 3 — Uniform quantile sampling for AQ-NBEATS. Standard AQ-NBEATS samples τ uniformly, giving equal training weight to all quantile levels. Tail quantiles (important for reserve planning) receive no additional emphasis. We investigate curriculum learning strategies to address this.

Gap 4 — No post-hoc calibration. Quantile models can be miscalibrated under seasonal distribution shift. CQR offers finite-sample coverage guarantees without retraining, but has not been applied to AQ-NBEATS.

Gap 5 — Architectural output rigidity. The basis-expansion output heads are quantile-independent; all quantile levels share the same intermediate representation. Richer quantile-conditioned output structures could improve distributional accuracy, but have not been evaluated on EMHIREs.

1.3.2 Contributions of This Study

C1 — Comprehensive ablation study. We conduct the most systematic evaluation of AQ-NBEATS configurations to date: 23 configurations, up to 7 independent random seeds, on the EMHIREs MHLV benchmark. This establishes a reliable ranking of enhancement strategies.

C2 — Exogenous feature integration. We demonstrate that weather and calendar features yield an approximate 20% CRPS reduction, directly addressing Gap 1 and confirming the hypothesis of Smyl et al. [16].

C3 — Curriculum learning protocol. We introduce a two-stage curriculum combining beta-distributed tail sampling with adaptive quantile scheduling, achieving the best multi-seed CRPS of 172.21 (addressing Gap 3).

C4 — Novel architectural prototypes. Three novel output head variants — QCBC, TCR, and DBE — are designed, implemented, and benchmarked as baselines for future research (addressing Gap 5).

C5 — CQR robustness validation. We empirically validate the robustness of CQR to seasonal distribution shift in a European multi-country electricity setting, achieving 96.0% empirical coverage (addressing Gap 4).

Chapter 2

Analysis and Specification of Requirements

This chapter identifies the key stakeholders of the probabilistic forecasting system, specifies both functional and non-functional requirements, and describes the domain constraints that shape the system design. The requirements are derived from the official project specifications document (PCD/26/02, ID 200) and from the analysis of the target application domain.

2.1 Requirement Analysis

2.1.1 Actors Identification

The system serves four distinct categories of stakeholders, classified by interaction frequency and operational criticality.

Primary Actors

1. **Energy Grid Operators.** These are the professionals responsible for real-time grid management, load balancing, and transmission security. They interact with the forecasting system on a continuous, daily basis. Their primary needs are 48-hour probabilistic demand forecasts updated every few hours, real-time dashboard displays, and automated capacity-breach alerts that trigger when the upper prediction interval quantile exceeds available generation capacity. For this actor, *reliability* and *timeliness* are the dominant requirements: a late or incorrect forecast can lead to reserve shortfalls or unnecessary balancing actions.
2. **Energy Traders.** These professionals buy and sell electricity in day-ahead, intraday, and balancing markets. They require multiple quantile levels simultaneously — not just a median — in order to price risk accurately and structure options or capacity agreements. They need the ability to specify custom quantile levels on demand, export forecast data in machine-readable formats (CSV, JSON), and compare forecasts across countries or time periods interactively. For traders, *flexibility* and *accuracy* are paramount.

Secondary Actors

1. **Energy Planners.** Strategic decision-makers responsible for long-term infrastructure planning (new generation capacity, grid reinforcement, interconnector investment). They interact with the system infrequently, accessing historical trend analyses, seasonality decompositions, and forward scenario reports. They require exportable PDF and Excel reports rather than real-time dashboard functionality.
2. **System Administrators.** Technical staff responsible for user account management, role-based access control, system configuration, and security audit logging. They do not interact with the forecasting models directly but are responsible for the integrity and security of the underlying platform.

2.1.2 Functional Requirements

Table 2.1 summarises the functional requirements mapped to each actor category.

Table 2.1: Actors and their functional requirements (from PCD/26/02).

Actor	Functional Requirements
Energy Grid Operators	(FR1) Generate 48-hour probabilistic forecasts with configurable quantile levels; (FR2) display real-time dashboard with automatic updates; (FR3) trigger capacity-breach alerts when predicted demand exceeds configurable thresholds; (FR4) visualise historical forecast-vs-actual overlays.
Energy Traders	(FR5) Generate multiple simultaneous quantile forecasts for risk assessment; (FR6) support custom quantile selection in $[0.001, 0.999]$; (FR7) export forecast data in CSV and JSON formats; (FR8) compare forecasts across countries or date ranges.
Energy Planners	(FR9) Generate long-term forecast scenarios with uncertainty bounds; (FR10) access trend analysis and seasonality reports; (FR11) export formatted PDF planning reports.
System Administrators	(FR12) Manage user accounts and assign roles; (FR13) configure system parameters (models, update frequency); (FR14) access audit logs and security event records.
All Users	(FR15) Upload and manage time series datasets (CSV); (FR16) automatic preprocessing (missing value imputation, outlier flagging, normalisation); (FR17) secure login with role-based feature access.

2.1.3 Non-Functional Requirements

Performance Requirements

- **Inference latency:** a 48-hour probabilistic forecast for one country at 100 quantile levels must be generated in under 5 seconds on the production hardware (NVIDIA A40, 24 GB VRAM). This is achievable with the AQ-NBEATS-OUT variant, whose inference cost is $O(1)$ in the number of quantiles requested.
- **Training throughput:** a full 15-epoch training run on all 35 EMHIRES countries must complete in under 10 hours on the A40 production environment. This is confirmed by our experiments (Section 5.1).
- **API response time:** the REST API must respond to forecast requests within 10 seconds under a load of up to 20 concurrent users.

Scalability Requirements

- The system must handle time series of heterogeneous lengths (spanning 2 to 13 years in the EMHIRES dataset) and sampling frequencies (currently hourly, potentially sub-hourly in future deployments).
- The training architecture must support adding new countries or regions without retraining from scratch, via the series embedding mechanism (Section 4.4).

Reliability Requirements

- **Fault tolerance:** automatic recovery from training failures via PyTorch Lightning checkpoint resumption.
- **Data integrity:** NaN values in test data are excluded from metric calculations rather than imputed, avoiding information leakage.
- **Quantile monotonicity:** the system must guarantee that delivered quantile forecasts are monotonically non-decreasing in τ (no quantile crossing). This is enforced by post-hoc sorting.

Security Requirements

- **Data protection:** electricity demand data, which may be commercially sensitive, is stored with AES-256 encryption at rest and TLS 1.3 encryption in transit.
- **Access control:** all endpoints are protected by JWT-based authentication; role-based access control prevents unauthorised access to administrative functions.
- **Audit logging:** all data access and configuration changes are logged with user ID, timestamp, and action description.

2.2 Needs of the Domain

2.2.1 Energy Domain Requirements

The electricity demand forecasting domain imposes constraints that must be embedded in both the data preprocessing and modelling layers:

- **Multiple temporal patterns:** European hourly electricity demand exhibits strong daily (24h), weekly (168h), and annual seasonality patterns. These patterns vary in shape and amplitude across countries and must be captured faithfully by the forecasting model.
- **Exogenous variable integration:** operational forecasting systems must incorporate weather forecast inputs (temperature, humidity, wind speed, pressure) and calendar information (public holidays, school calendars) that are available 48+ hours ahead. The AQ-NBEATS baseline ignores these drivers, creating a systematic underestimation of demand extremes.
- **Demand peak management:** peak demand events — driven by cold spells, heatwaves, or unusual calendar patterns — represent the highest-risk scenarios for grid operators. The forecasting model must produce calibrated tail quantiles ($\tau \geq 0.90$) to support reserve planning for these events.
- **Operational forecast horizon:** the 48-hour forecast horizon is imposed by European power market regulations, which require day-ahead demand declarations submitted 12–36 hours before delivery.

2.2.2 Time Series Domain Requirements

- **Multi-series management:** the system must handle 35 heterogeneous country series with demand levels ranging from 383 MWh (Montenegro) to 58,941 MWh (Germany). The normalised MQN loss (1.10) provides scale invariance, but series-level embeddings are needed to capture country-specific biases.
- **Missing value handling:** the EMHIRES dataset contains missing values (NaN) primarily at year-end periods for Estonia, Italy, and Latvia. The training pipeline applies 168h forward-fill (one-week lag); test NaNs are excluded from metrics.
- **Timestamp consistency:** all series are indexed by UTC timestamp, and the preprocessing pipeline enforces strict temporal alignment before constructing sliding-window samples.

2.3 Requirements Specification

2.3.1 Modelling Language

All neural architectures are specified in Python 3.10 using PyTorch 2.0+. Experimental configurations are managed via YAML files (one file per experiment variant), parsed at runtime by OmegaConf. Training is orchestrated by PyTorch Lightning 1.8+. Metrics

are computed in Python/NumPy and logged in structured CSV files (one row per epoch per seed). Visualisations are produced by Matplotlib and Plotly.

The modelling language choice ensures reproducibility: every experiment can be re-run exactly by specifying its YAML config and random seed.

2.3.2 Use Case Diagram

The system’s six primary use cases, organised by actor, are:

1. **Authenticate and access (All Users)**: users log in via secure credentials; the system grants feature access according to actor type. JWT tokens expire after 24 hours; refresh tokens extend sessions.
2. **Upload and preprocess time series (All Users)**: users upload CSV data; the system validates format, detects and flags outliers, applies missing value imputation (168h forward-fill for training), and normalises series for display.
3. **Configure and run forecast (Grid Operators, Traders)**: users select model configuration (backbone variant, quantile levels, forecast horizon); the system loads the appropriate checkpoint and generates probabilistic forecasts within the guaranteed latency SLA.
4. **Visualise and export results (Grid Operators, Traders, Planners)**: users view interactive Plotly.js prediction interval charts; export quantile forecasts as CSV or PDF summaries.
5. **Monitor and alert (Grid Operators)**: the system continuously compares the upper PI bound against user-defined capacity thresholds and generates alerts via email or dashboard notification.
6. **Administer system (System Administrators)**: management of user accounts, role assignments, model checkpoint updates, and audit log review.

2.3.3 Sequence Diagrams

Forecast generation sequence (nominal path):

1. User authenticates via POST /auth/login; server returns JWT token.
2. Frontend sends POST /forecast with body {country, horizon=48, quantiles=[0.025, 0.5, 0.975]}.
3. Backend validates JWT; resolves model checkpoint from configuration store.
4. `AQNBEATSModel.predict()` generates quantile forecasts for the 48-hour horizon in a single forward pass (AQ-OUT variant: $O(1)$ in number of quantile levels).
5. Backend stores results in PostgreSQL; pushes JSON response to frontend.
6. Frontend renders interactive Plotly.js PI chart.
7. If any forecast value exceeds the configured capacity threshold, the alert service triggers a notification.

Training sequence (researcher workflow):

1. Researcher selects YAML config (backbone, exogenous features, curriculum stages, random seeds).
2. `EMHIRESDataModule.setup()` loads and preprocesses the full ENTSO-E CSV; constructs train/validation/test splits.
3. `Trainer.fit()` runs the training loop for 15 epochs; callbacks save the best checkpoint by validation CRPS.
4. `Trainer.test()` loads best checkpoint; computes CRPS, N-CRPS, coverage, MAE, MAPE, SMAPE on the 2018 test set.
5. For multi-seed experiments, steps 3–4 are repeated for each seed; results are aggregated as mean $\pm$ std.

Chapter 3

Design

This chapter describes the architectural and software design decisions that underpin the experimental system. We adopt a Model-View-Controller (MVC) decomposition adapted to the ML research context, describe the physical computing infrastructure, and detail the object-level design through sequence, activity, package, and class diagrams.

3.1 MVC Model

The MVC paradigm partitions the system into three loosely coupled layers, each with a clearly defined responsibility:

Model layer. All neural network components live in the Model layer: the NBEAT-SAQCAT backbone, exogenous encoder modules, series embedding lookup tables, novel output heads (QCBC, DBE), and the CQR calibrator. Each component is implemented as a standalone PyTorch `nn.Module`, enabling unit testing, independent composition, and serialisation via PyTorch’s checkpoint mechanism. The Model layer also includes the data pipeline — `EMHIRESDataModule` and the custom PyTorch `Dataset` — which are responsible for loading, preprocessing, and batching training samples.

View layer. In this research context, the View layer corresponds to the human-readable outputs of the system: structured metric CSV logs (one row per epoch per seed), comparison tables, training curve plots, and quantile forecast visualisations. These outputs are the primary interface between the system and its users (the research team and the jury). In the web application deployment, the View layer is implemented as a React/TypeScript frontend rendering interactive Plotly.js charts.

Controller layer. YAML configuration files and the PyTorch Lightning `Trainer` serve as the Controller. YAML files define all hyperparameters (architecture, loss, training schedule, seeds), making experiments declarative, reproducible, and diffable in version control. The `Trainer` orchestrates the full training lifecycle: data loading, gradient updates, learning rate scheduling, checkpoint saving, and test evaluation. The REST API backend acts as a secondary controller layer in the web application deployment.

3.2 Physical Architecture

Three distinct computing environments were used across development, training, and production deployment, as specified in PCD/26/02.

Table 3.1: Hardware configuration used in the project.

Environment	Specification
Machine 1 (Development)	Intel Core i7-10700K / AMD Ryzen 7 3700X (8 cores, 3.8 GHz); 16–32 GB DDR4 RAM; 500 GB NVMe SSD; NVIDIA RTX 3070 (8 GB VRAM); CUDA 11.8; Docker containerised environment. Used for initial architecture prototyping, unit tests, and single-seed configuration sweeps in fp16 mixed precision.
Machine 2 (OMENCVteam)	HP OMEN 40L Gaming Desktop; 13th Gen Intel Core i7-13700K (24 threads, 3.4 GHz); 32 GB DDR5 RAM; 2 TB NVMe SSD; NVIDIA GeForce RTX 4070 Ti (12 GB VRAM); Ubuntu 22.04.5 LTS; TeamViewer remote access. Used for intermediate-scale experiments (5 seeds) requiring more VRAM than Machine 1.
Virtual Machine (CCK)	64 vCPUs; 256 GB DDR4 ECC RAM; 1 TB SSD; NVIDIA A40-1Q (24 GB VRAM) — primary environment for all multi-seed production training runs in fp32 full precision. Training throughput: approximately 13 iterations/second with batch size 2048.

Mixed-precision (fp16) training was used on the RTX 3070 to fit the exogenous model (larger input dimension) within the 8 GB VRAM budget. Full precision (fp32) was mandatory on the A40 to ensure numerical stability in production experiments, particularly for the CQR calibration which requires precise quantile estimation.

3.3 Detailed Conception

3.3.1 Object Sequence Diagram

The forward pass of the full AQ-NBEATS model with exogenous features and series embeddings proceeds as follows at the object level:

1. `EMHIRESDataModule.__getitem__()` returns a tuple $(x_s, E_s^{\text{hist}}, E_s^{\text{fore}}, \text{sid}_s, y_s)$ comprising demand history, historical and future exogenous features, series identifier, and forecast target.
2. `ExogEncoder.forward()` encodes $[E_s^{\text{hist}}; E_s^{\text{fore}}] \in \mathbb{R}^{(W+H) \times d_e}$ via a two-layer MLP with ReLU activations, producing $\mathbf{z}_s^{\text{exog}} \in \mathbb{R}^W$.
3. `SeriesEmbedding.forward()` looks up the embedding $\mathbf{e}_s \in \mathbb{R}^{d_e}$ for country s and projects it to $\mathbb{R}^W$ via a linear layer with small initialisation (std 0.01).
4. The combined input $\tilde{x}_s = x_s + \mathbf{z}_s^{\text{exog}} + P(\mathbf{e}_s)$ is concatenated with τ to form $[\tilde{x}_s; \tau] \in \mathbb{R}^{W+1}$.

5. `NBEATSAQCAT.forward()` passes this through 30 residual blocks; the final forecast $\hat{q}_\tau \in \mathbb{R}^H$ is the sum of block outputs.
6. `MQNLoss.forward()` computes the normalised pinball loss (1.10) and returns the scalar loss value for backpropagation.

3.3.2 Activity Diagram

The complete experiment lifecycle for a multi-seed production run is:

1. Parse YAML configuration file; validate all required fields and type-check hyperparameters.
2. Instantiate experiment variant (select backbone class, loss function, exogenous encoder, sampling strategy).
3. For each seed $s \in \{1, \dots, N_{\text{seeds}}\}$:
 - a. Set global random seed via `pl.seed_everything(s)`.
 - b. Instantiate model with freshly randomised weights.
 - c. Call `Trainer.fit()`:
 - For each epoch: iterate over batches, compute forward pass, backpropagate, update with AdamW.
 - `ModelCheckpoint` saves best checkpoint by validation CRPS.
 - Learning rate scheduler (inverse square-root with warm-up) updates LR after each batch.
 - d. Call `Trainer.test()` on the best checkpoint: compute CRPS, N-CRPS, coverage, MAE, MAPE, SMAPE for all 35 countries.
 - e. Append results row to experiment CSV log.
4. Aggregate: compute mean $\pm$ std across all seeds.
5. Write final summary row to master comparison CSV.

3.3.3 Package Diagram

The codebase is organised into four top-level Python packages with clear dependency arrows:

- **data**: EMHIRE dataset loading, missing-value imputation, feature engineering, and the PyTorch Lightning `DataModule`. Has no dependencies on **models** or **losses**.
- **models**: all backbone classes (`NBEATSAQCAT`, `NBEATSAQFILM`), exogenous encoders, series embedding modules, and novel output heads (`NBEATSAQQCBC`, `NBEATSDBEHead`). Depends only on **losses** for the training objective.
- **losses**: normalised pinball (MQN), smooth Huber pinball, hybrid CRPS-pinball, and monotonicity penalty. Has no external dependencies.
- **evaluation**: CRPS computation, coverage calculation, the `CQRCalibrator`, and metric aggregation utilities. Depends on **models** for inference.

3.3.4 Class Diagram

The core class hierarchy, with inheritance and composition relationships:

- **NBEATSBLOCK** — single residual block with L fully-connected layers and separate backcast and forecast projection heads. Base component for all backbone variants.
- **NBEATSAQCAT** (inherits **LightningModule**) — 30-block stack with CAT quantile injection; 74.1M parameters. The primary baseline backbone.
- **NBEATSAQFILM** (inherits **LightningModule**) — FiLM-conditioned variant; 71.7M parameters. Every block contains an additional FiLM layer.
- **ExogEncoder** (composed by **NBEATSAQCAT**) — two-layer MLP encoder for continuous weather features plus embedding tables for calendar features. Output dimension: W .
- **SeriesEmbedding** (composed by **NBEATSAQCAT**) — learnable embedding table of size $N_{\text{countries}} \times d_e$ (35×32). Linear projection to $\mathbb{R}^W$.
- **NBEATSAQQCBC** (inherits **NBEATSAQCAT**) — modulates each block’s basis coefficients by τ via learned scalar gates $\theta_k(\tau)$.
- **NBEATSDBEHead** (composed by **NBEATSAQCAT**) — 3-component Laplace mixture output head blended with the baseline forecast via a learned mixing weight.
- **CQRCalibrator** — standalone calibration module. Accepts a fitted quantile model and a calibration split; computes and stores the conformity score quantile Δ . Wraps any backbone without modifying its weights.

Chapter 4

Methodology of the Study

This chapter describes the complete experimental methodology, from dataset acquisition and preprocessing through model construction and training strategy, to evaluation protocols and the multi-seed reproducibility framework.

4.1 Data Collection and Understanding

4.1.1 The EMHIRES Dataset

The dataset is sourced from the ENTSO-E Transparency Platform (www.entsoe.eu/data/power-stats) and is publicly available alongside the model code at <https://github.com/boreshkinai/any-quantile> [16]. It comprises hourly electricity demand measurements for 35 European countries over the 13-year period from 2006 to 2018, totalling approximately 4 million individual observations.

The dataset exhibits several characteristics that make it both practically representative and challenging for forecasting:

- **Heterogeneous demand levels.** Average hourly demand ranges from 383 MWh (Montenegro) to 58,941 MWh (Germany), a factor of 154. This heterogeneity motivates the use of the scale-invariant normalised MQN loss and the exploration of per-country series embeddings.
- **Multiple seasonalities.** Each series exhibits strong daily (24h), weekly (168h), and annual seasonal patterns. Daily patterns reflect residential and commercial activity cycles; weekly patterns reflect reduced industrial activity on weekends; annual patterns reflect heating and cooling loads.
- **Non-stationary trends.** Several countries show long-term demand growth (developing economies) or decline (industrial restructuring, energy efficiency programmes) over the 13-year observation window.
- **Heterogeneous variability.** The daily variation coefficient ($v_d = 100s/\bar{z}$, where s and $\bar{z}$ are the daily standard deviation and mean of hourly demand) exceeds 17% for Albania, Italy, Latvia, and Great Britain. Iceland exhibits the smallest variation overall. Norway, France, Sweden, and Macedonia show the highest yearly variability ($v_y > 18\%$), reflecting strong heating-degree-day sensitivity.
- **Missing values.** NaN values are present in the final months of 2018 for Estonia, Italy, and Latvia. These are handled differently in training (168h forward-fill) and testing (exclusion from metrics).

4.1.2 Exogenous Weather Features

For exogenous-augmented experiments, we construct a feature matrix $\mathbf{E} \in \mathbb{R}^{T \times 8}$ comprising four continuous meteorological features and four calendar embeddings:

Continuous meteorological features:

1. Temperature ($^{\circ}\text{C}$) — capital city representative.
2. Relative humidity (%).
3. Atmospheric pressure (hPa).
4. Wind speed (m/s) at 10m height.

Calendar features (categorical, embedded):

1. Hour of day (0–23), embedding dimension 4.
2. Day of week (0–6), embedding dimension 4.
3. Month of year (1–12), embedding dimension 4.
4. Weekend indicator (0/1), embedding dimension 2.

Continuous features are standardised over the training period (mean 0, standard deviation 1). Calendar features use learnable embedding tables initialised by the standard normal distribution. This allows the model to learn non-linear, data-driven representations of temporal patterns rather than relying on one-hot encoding or hand-crafted Fourier features.

A key advantage of incorporating future calendar features at training time is that they are perfectly known at inference time (the future calendar is deterministic), eliminating forecast uncertainty for this signal. Future weather forecasts introduce some uncertainty; in our experimental setup we use reanalysis data as a proxy for operational weather forecasts.

4.2 Dataset Organisation

4.2.1 Data Splits

Following Smyl et al. [16] exactly for reproducibility:

- **Training set:** 2006-01-01 to 2016-12-31 (11 years).
- **Validation set:** 2017-01-01 to 2017-12-31 (1 year).
- **Test set:** 2018-01-01 to 2018-12-31 (1 year).
- **CQR calibration / CQR test:** Jan–Jun 2018 / Jul–Dec 2018 (seasonal split for distribution-shift testing).

Hyperparameters are tuned on the validation set. The final model is retrained on the combined train+validation set before test evaluation. This ensures that the test set is never seen during model selection.

4.2.2 Sliding-Window Sample Construction

Each training sample is a tuple: $(x_s, \mathbf{E}_s^{\text{hist}}, \mathbf{E}_s^{\text{fore}}, \text{sid}_s, y_s)$, where:

- $x_s \in \mathbb{R}^W$ is the lookback window of length $W = 168$ hours (7 days) of normalised demand.
- $\mathbf{E}_s^{\text{hist}} \in \mathbb{R}^{W \times d_e}$ is the historical exogenous feature matrix for the same window.
- $\mathbf{E}_s^{\text{fore}} \in \mathbb{R}^{H \times d_e}$ is the future exogenous feature matrix for the forecast horizon $H = 48$ hours.
- $\text{sid}_s \in \{0, \dots, 34\}$ is the country index.
- $y_s \in \mathbb{R}^H$ is the forecast target.

Samples are constructed with a stride of 24 hours for training and evaluated without overlap for validation and test. All 35 country series contribute to each batch, providing automatic cross-country learning.

4.3 Data Preprocessing

The preprocessing pipeline is fixed across all experiments to ensure fair comparison:

1. **Missing value imputation:** each NaN in the training set is replaced by the value 168 hours prior (one full week). This exploits the strong weekly seasonality and avoids information leakage from future values.
2. **Input scaling:** for the NBEATSAQCAT backbone, no global normalisation is applied to the demand history (per [16]). The MQN loss provides implicit scale invariance.
3. **Exogenous feature scaling:** continuous weather features are standardised using training-set statistics only.
4. **Calendar feature encoding:** integer indices passed to embedding lookup tables; no further preprocessing needed.
5. **Test set NaN exclusion:** observations at timestamps corresponding to NaN values in Estonia, Italy, and Latvia are excluded from all metric calculations.

4.4 Model Building

4.4.1 Baseline Configuration

The AQ-NBEATS baseline configuration, matching [16]:

- Backbone: NBEATSAQCAT (quantile level concatenated to input).
- Residual blocks: $R = 30$.
- Hidden layers per block: $L = 3$.

- Hidden dimension: $d_h = 1024$.
- Lookback window: $W = 168$ (7 days).
- Forecast horizon: $H = 48$ (2 days).
- No dropout, no weight decay.
- Total parameters: 74.1M.

4.4.2 Exogenous Feature Encoder

The exogenous encoder processes both historical and future weather features:

1. Calendar embeddings are looked up and projected to $\mathbb{R}^{d_{\text{cal}}}$ by a linear layer.
2. Continuous weather features are passed through a two-layer MLP: input dimension 4, hidden dimension 64, output dimension d_{proj} .
3. The concatenated continuous and calendar representations are projected to $\mathbb{R}^W$ for historical features and $\mathbb{R}^H$ for future features, and added to the corresponding demand sequences.

4.4.3 Series Embedding Module

A learnable embedding table of size $35 \times d_e$ (with $d_e = 32$) provides per-country representations. At each forward pass, the embedding $\mathbf{e}_s$ for country s is projected to $\mathbb{R}^W$ by a linear layer P and added to the input as an additive bias:

$$\tilde{x}_s = x_s + \alpha \cdot P(\mathbf{e}_s), \quad (4.1)$$

where α is a learnable scalar initialised to 0.1 and P has weights initialised with standard deviation 0.01. The small initialisation ensures that embeddings begin as near-zero perturbations, allowing the backbone to dominate early in training and preventing the embeddings from overwhelming the exogenous signal in combined configurations.

4.4.4 Novel Architectural Contributions

Three novel output head variants are designed and evaluated in this study. Each replaces or augments the standard linear forecast projection $\mathbf{F}^r \mathbf{h}^{r,L}$ in the final or all blocks.

QCBC (Quantile-Conditioned Basis Coefficients): modulates each block’s basis expansion coefficients by τ :

$$\theta_k(\tau) = \theta_k^{\text{base}} \cdot (1 + \alpha_k \cdot g(\tau)), \quad (4.2)$$

where $g(\tau)$ is a small MLP mapping the scalar τ to a K -dimensional modulation vector, α_k are learnable modulation scales, and θ_k^{base} are the standard basis coefficients. The modulation scale is initialised to 0.001 to ensure near-identical behaviour to the baseline at the start of training.

TCR (Temporal Coherence Regularisation): adds a loss term penalising large temporal differences in quantile forecasts:

$$\mathcal{L}_{\text{TCR}} = \lambda_{\text{TCR}} \sum_{h=1}^{H-1} (\hat{q}_{\tau}(h+1) - \hat{q}_{\tau}(h))^2, \quad (4.3)$$

encouraging smooth forecast trajectories. The regularisation weight λ_{TCR} is a hyper-parameter tuned on the validation set.

DBE (Distributional Basis Expansion): replaces the forecast head with a 3-component Laplace mixture model:

$$p(y|\mathbf{h}, \tau) = \sum_{k=1}^3 \pi_k(\mathbf{h}) \cdot \text{Laplace}(y; \mu_k(\mathbf{h}), b_k(\mathbf{h})), \quad (4.4)$$

where π_k , μ_k , and b_k are predicted by a linear layer from the block hidden state. The quantile $\hat{q}_{\tau}$ is computed numerically from the predicted mixture CDF. The final output blends the DBE quantile with the standard basis-expansion forecast:

$$\hat{q}_{\tau}^{\text{final}} = (1 - \alpha)\hat{q}_{\tau}^{\text{base}} + \alpha\hat{q}_{\tau}^{\text{DBE}}, \quad (4.5)$$

where $\alpha = \sigma(-10) \approx 0$ at initialisation ensures training stability.

4.5 Training Strategy

4.5.1 Optimizer and Schedule

All experiments use the AdamW optimiser with the following settings:

- Peak learning rate: 0.0005 for baseline and exogenous experiments; 0.0001 for novel architectural heads (conservative, to avoid instability from large parameter count changes).
- Learning rate schedule: inverse square-root with a warm-up period of 400 batches, then decay proportional to $1/\sqrt{t}$.
- Batch size: 1024 for all NBEATSAQCAT experiments on the A40; 512 for exogenous experiments on the RTX 3070 (memory constraint).
- Training duration: 15 epochs for all experiments.

4.5.2 Quantile Sampling Strategies

Three sampling strategies are evaluated for τ in training:

Uniform sampling (baseline): $\tau \sim U(0, 1)$ for each training sample. Each quantile level receives equal gradient weight over a full epoch.

Beta sampling: $\tau \sim \text{Beta}(\beta_0, \beta_0)$ with $\beta_0 = 0.3$. This distribution is U-shaped, over-representing tail quantiles (τ near 0 or 1) relative to the uniform. Motivated by the AQ-ESRNN design choice, which demonstrated that tail over-sampling benefits CRPS even for models without saturating activations.

Adaptive sampling: a dynamic scheme that focuses training on quantile levels where the current model is most miscalibrated. Specifically, coverage is estimated on a rolling mini-validation set; quantile levels with the largest absolute coverage gap are sampled with proportionally higher probability. The adaptation temperature controls how aggressively the distribution is skewed.

4.5.3 Curriculum Learning Protocols

Two curriculum protocols are evaluated:

Curriculum v1: progressive feature stacking in three stages. Stage 1 trains on demand + exogenous features with uniform sampling. Stage 2 adds series embeddings. Stage 3 switches to adaptive sampling. Each stage resumes from the previous checkpoint.

Curriculum v2: sampling-first curriculum. Stage 1 uses beta sampling to establish wide, calibrated intervals. Stage 2 switches to adaptive sampling to sharpen the distribution. Optionally, Stage 3 adds series embeddings. Based on our experiments, Stage 2-v2 (without Stage 3) produces the best overall results.

4.5.4 Multi-Seed Evaluation Protocol

All key experimental configurations are evaluated with multiple independent random seeds (varying network initialisation and training data order). Results are reported as mean $\pm$ standard deviation across seeds. Single-seed results are clearly labelled and interpreted with appropriate caution. The production AQ-NBEATS baseline in the paper uses 8 seeds with median ensembling; our reproduced baseline uses 7 seeds and matches the published result within 10%.

4.6 Evaluation Metrics and Protocol

Test evaluation computes the following metrics on the 2018 holdout set:

- **CRPS** (Eq. 1.2): primary metric, 100 quantile levels, averaged over all countries and horizons.
- **N-CRPS** (Eq. 1.4): normalised CRPS for cross-country comparison.
- **Coverage-0.95** (Eq. 1.5): empirical coverage of the 95% PI.
- **MAPE**: mean absolute percentage error on the median forecast.
- **MAE, SMAPE**: supplementary point forecast metrics.

All metrics exclude NaN-marked test observations for Estonia, Italy, and Latvia in their final months of 2018.

Chapter 5

Implementation and Results

This chapter details the software and hardware environments used, presents the full experimental results across all twenty-three configurations, analyses the findings, and discusses their implications for probabilistic electricity demand forecasting.

5.1 Environment and Working Tools

5.1.1 Hardware Environment

Three computing environments were used as specified in Chapter 3 (Table 3.1). Machine 1 (RTX 3070, 8 GB VRAM) served as the primary development and prototyping environment. Mixed-precision fp16 training enabled the exogenous feature model (which requires additional VRAM for the feature encoders) to fit within the 8 GB budget. Machine 2 (RTX 4070 Ti, 12 GB VRAM) provided intermediate-scale validation of key configurations before committing to multi-seed production runs. The CCK virtual machine (A40, 24 GB VRAM) was the sole environment for all production multi-seed experiments, ensuring numerical precision and experiment reproducibility.

Training time summary:

- Baseline NBEATSAQCAT (A40, fp32, batch 2048): 8 hours per full 15-epoch run per seed.
- Exogenous variants (A40, fp32, batch 1024): approximately 14 hours per seed due to larger batch size reduction and exogenous encoder overhead.
- Exogenous variants (RTX 3070, fp16, batch 512): 20–24 hours per seed.
- Novel head variants (A40, fp32): 8–10 hours per seed.

5.1.2 Software Environment

Programming Languages and Core Libraries

Python 3.10 was used throughout, providing a stable ecosystem for scientific computing (NumPy, SciPy, Pandas), deep learning (PyTorch), and data visualisation (Matplotlib, Plotly).

Deep Learning Framework

- **PyTorch 2.0+**: automatic differentiation engine, CUDA kernel dispatch, and tensor operations. The compiled model mode (`torch.compile`) was tested but disabled in production due to compatibility issues with PyTorch Lightning callbacks.
- **PyTorch Lightning 1.8+**: high-level training orchestration providing epoch loops, gradient accumulation, multi-GPU support, automatic mixed precision (AMP), and the `ModelCheckpoint` callback for best-epoch model selection.
- **OmegaConf / Hydra**: type-safe YAML configuration parsing, enabling hierarchical experiment configuration with CLI overrides for hyperparameter sweeps.

Data and Evaluation Stack

- **NumPy / SciPy / Pandas**: data loading, preprocessing, and metric computation (CRPS, coverage, MAPE).
- **GluonTS**: utility functions for time series batching and evaluation; used for baseline model implementations (DeepAR, WaveNet, TFT, Transformer, MLP).
- **scikit-learn**: used for the QR baseline (`QuantileRegressor`) and preprocessing utilities.

Web Application Stack

The forecasting system is delivered as a full-stack web application for deployment in operational grid management contexts:

Frontend:

- React 18+ with TypeScript for type-safe, component-based UI.
- Next.js for server-side rendering and file-based routing.
- TailwindCSS for responsive, utility-first styling.
- Plotly.js for interactive probabilistic forecast visualisations with zoomable prediction interval bands.

Backend:

- Python FastAPI for high-performance asynchronous REST endpoints.
- PyTorch inference engine serving pre-loaded AQ-NBEATS checkpoints; single-pass forward function handles arbitrary quantile level sets.
- Asynchronous request handling supporting up to 20 concurrent forecast requests within the 10-second SLA.

Database and Storage

- **PostgreSQL 15+**: primary relational store for time series data, user records, forecast results, and audit logs. Tables are indexed on (country, timestamp) for efficient historical queries.
- **Redis**: in-memory cache for real-time dashboard updates and asynchronous training job queue (via Celery workers).
- **CSV logs**: all experiment results stored in structured CSV files (`exp_name`, `seed`, `epoch`, `crps`, `ncrps`, `coverage`, `mape`, `mae`, `smape`) for post-hoc analysis and reproducibility.

Development and DevOps Tools

- **Git / GitHub**: version control with a branch-per-experiment strategy ensuring that each configuration change is tracked.
- **Docker / Docker Compose**: containerised environment eliminating dependency conflicts across the three machines.
- **TensorBoard**: real-time training curve visualisation (loss, learning rate, validation CRPS by country).
- **Jupyter Notebooks**: interactive result analysis, visualisation of PI charts, and ablation study plots.
- **VS Code**: primary IDE with Python, Pylance, and YAML extensions.

5.2 Computational Results

5.2.1 Baseline Reproduction

Experiment 3 reproduces the AQ-NBEATS baseline using the published hyperparameters [16]. With 7 random seeds, we obtain $\text{CRPS} = 190.78 \pm 1.42$, $\text{MAPE} = 2.376\%$, and $\text{coverage-0.95} = 0.942 \pm 0.011$. The CRPS is approximately 9.6% below the published 211.22, consistent with expected variance in the 8-seed ensemble reduction (the published value uses median ensembling of 8 models; our evaluation uses individual seeds). The coverage value of 0.942 is slightly below the 0.95 nominal target, indicating mild under-coverage in the baseline, which CQR will later correct.

5.2.2 Exogenous Feature Augmentation

Exogenous features alone (Exp 20) achieves the single largest improvement found in this study: $\text{CRPS} = 168.46$, a $\approx 20.3\%$ reduction compared to the published AQ-NBEATS baseline of 211.22. This result was obtained with a single seed; multi-seed variance may be substantial. Three training adjustments from the baseline were necessary for convergence:

- Learning rate reduced from 0.0005 to 0.0001 to accommodate the additional feature dimensions without destabilising early training.

- Batch size reduced from 1024 to 512 due to VRAM constraints on the development machine (fp16 training).
- Early stopping at epoch 12 of 15 (validation CRPS plateaued).

Series embeddings alone (Exp 19) achieves $\text{CRPS} = 206.51 \pm 0.88$ across 7 seeds, a modest 2.2% improvement over the published baseline. The 32-dimensional per-country embeddings capture demand level biases between countries but provide no direct access to the meteorological drivers of demand variability. Coverage improves slightly to 0.936, suggesting that country-specific calibration is marginally better.

Combined exogenous features + series embeddings (Exp 21) achieves $\text{CRPS} = 174.09 \pm 0.95$ with 5 seeds. The embedding scale parameter was reduced from 0.15 to 0.08 in this configuration to prevent the low-dimensional country embedding from competing with the richer exogenous signal. Coverage drops to 0.912, likely because the exogenous model produces sharper (less conservative) intervals in regions where temperature reliably predicts demand.

Combined with monotonicity penalty (Exp 22) failed to improve upon Exp 21: $\text{CRPS} = 239.22$, coverage = 0.985. The soft non-crossing penalty (5 quantiles per sample, weight 0.03) overwhelmed the exogenous training signal, forcing the model to produce excessively wide intervals to minimise crossing violations. This interaction between the monotonicity constraint and the richer feature space is an important negative finding for future practitioners.

Table 5.1: Exogenous feature and series embedding experiment results. CRPS lower is better; coverage target is 0.95.

Exp	Configuration	Seeds	CRPS	Cov-0.95	MAPE (%)	ΔCRPS
Paper	AQ-NBEATS (paper)	8	211.22	—	2.47	—
Exp 3	Baseline reproduced	7	190.78 ± 1.42	0.942 ± 0.011	2.38	−9.6%
Exp 19	Series embed only	7	206.51 ± 0.88	0.936 ± 0.002	2.44	−2.2%
Exp 20	Exog only (1 seed)	1	168.46	0.919	2.11	−20.3%*
Exp 21	Exog + Series	5	174.09 ± 0.95	0.912 ± 0.009	2.13	−17.6%
Exp 22	Exog+Series+Monotone	5	239.22	0.985	2.89	+13.2%

*Single seed; multi-seed mean may differ.

5.2.3 Curriculum Learning and Quantile Sampling

Adaptive Sampling Without Curriculum

Experiments 23–25 applied adaptive quantile sampling directly on top of the Exog+Series configuration (Exp 21), without any curriculum ordering. All three variants degraded performance (Table 5.2), with coverage dropping to 0.866–0.874, well below the 0.95 target. This negative result reveals an important finding: adaptive sampling is only beneficial when the model already has reasonable tail coverage as a starting point. Without it, the adaptive scheme over-concentrates on the worst-performing quantile levels, which may change erratically during early training, causing gradient instability.

Table 5.2: Effect of adaptive quantile sampling applied directly on Exp 21. All multi-seed, 5 seeds.

Configuration	CRPS	Cov-0.95	Status
Exog+Series (Exp 21) reference	174.09±0.95	0.912±0.009	Reference
+Adaptive (5q, temp=0.8)	177.60	0.866	Worse
+Adaptive (3q, temp=1.0)	178.03	0.874	Worse
+Adaptive (2q, temp=1.2, scale=0.05)	177.53	0.868	Worse

Curriculum v1: Feature-Staging

Curriculum v1 builds the model progressively across three stages, each resuming training from the previous checkpoint: Stage 1 (exogenous only) achieves CRPS 173.0; Stage 2 adds series embeddings, paradoxically increasing CRPS to 181.0; Stage 3 adds adaptive sampling, further degrading to 190.0 with poor coverage (0.885). The deterioration in Stage 2 suggests that adding embeddings on top of a trained exogenous model requires a warm-up period that the fixed-epoch training schedule does not provide.

Curriculum v2: Sampling-First

Curriculum v2 reverses the priority, establishing tail calibration before sharpening the distribution:

- **Stage 1** (Exog + Beta sampling): initialises the model with a distribution biased toward tail quantiles. Achieves CRPS 183.04 with coverage 0.947 — nearly meeting the 0.95 target from the first stage.
- **Stage 2** (Exog + Adaptive sampling): resumes from Stage 1 and focuses on regions of miscalibration. Achieves the **best multi-seed result: CRPS 172.21, coverage 0.907**, representing an 18.5% improvement over the published baseline.
- **Stage 3** (Exog + Adaptive + Series): adding series embeddings after curriculum training slightly degrades CRPS to 172.75 and reduces coverage to 0.895, confirming Stage 2 as the optimal endpoint.

Table 5.3: Curriculum learning stage results. Best configuration in bold.

Stage	Configuration	CRPS	Cov-0.95
<i>Curriculum v1 (feature-staging, 3 seeds)</i>			
Stage 1	Exog (baseline)	173.0	—
Stage 2	Exog + Series	181.0	—
Stage 3	Exog + Series + Adaptive	190.0	0.885
<i>Curriculum v2 (sampling-first, 5 seeds)</i>			
Stage 1	Exog + Beta sampling	183.04±1.9	0.947±0.012
Stage 2	Exog + Adaptive (best)	172.21±1.5	0.907±0.008
Stage 3	Exog + Adaptive + Series	172.75±1.8	0.895±0.011

The key insight from v2 is that beta sampling in Stage 1 establishes well-calibrated tail behaviour first, and adaptive sampling in Stage 2 then compresses the distribution toward better CRPS without sacrificing too much coverage. The 0.907 coverage in Stage 2-v2 falls below 0.95, but CQR (Section 5.4) corrects this in post-processing.

5.2.4 Novel Architectural Contributions

All three novel output heads are evaluated on the standard NBEATSAQCAT backbone without exogenous features, using the same hyperparameters as the baseline to isolate the architectural effect (Table 5.4).

Table 5.4: Novel architectural head results (5 seeds unless noted). Baseline comparison: published AQ-NBEATS CRPS = 211.22.

Contribution	Architecture	CRPS	Cov-0.95	Params
Baseline (paper)	NBEATSAQCAT	211.22	—	74.1M
QCBC	NBEATSAQQCBC	213.22±2.65	0.927±0.008	74.1M
TCR	NBEATSAQCAT + TCR	217.44±5.42	0.923±0.016	74.1M
DBE	NBEATSAQCAT + DBE	212.86±1.50	0.924±0.015	75.1M

QCBC modulates each block’s basis coefficients by τ (Eq. 4.2). The near-baseline CRPS (213.22 vs. 211.22) suggests that the standard basis-expansion output already provides sufficient quantile representational capacity within the 74.1M parameter budget. The modulation scale of 0.001 was chosen to prevent training instability; larger values led to divergence in preliminary experiments.

TCR penalises large temporal differences in quantile forecasts. The higher CRPS (217.44) and largest standard deviation (5.42) reflect instability in the adaptive penalty weight. Fixed-weight configurations (tested but not shown) achieved 214–216, still below the baseline. The fundamental tension is that temporal smoothness and CRPS sharpness are competing objectives for electricity demand, which exhibits sharp morning and evening demand ramps.

DBE replaces the linear forecast head with a 3-component Laplace mixture model, providing a richer representation of multi-modal forecast distributions. DBE achieves $\text{CRPS} = 212.86$ with a best-seed result of 210.99 — marginally below the paper baseline. The blend weight α (Eq. 4.5) converged to approximately 0.08 across seeds, suggesting the mixture head contributes a modest corrective signal on top of the basis-expansion forecast. The 937K additional parameters (the mixture MLP heads) add limited value on the 35-country EMHIREs dataset, which may lack sufficient diversity for the mixture to learn distinct distributional modes.

5.2.5 Alternative Loss Functions and Non-Crossing Constraints

Table 5.5 shows that all alternative loss functions underperform the MQN baseline (1.10).

Table 5.5: Alternative loss function results (single seed unless noted).

Method	Backbone	CRPS	Cov-0.95
MQN (baseline)	NBEATSAQCAT	211.22	—
Smooth Huber pinball	NBEATSAQCAT	220.34	0.898
Smooth Arctan pinball	NBEATSAQFILM	249.91	0.916
Hybrid CRPS-Pinball	NBEATSAQFILM	250.73	0.913
Non-crossing (cumsum)	NBEATSAQCAT	217.06	0.922

The smooth Huber variant (220.34) replaces the sharp kink of the standard pinball loss with a smooth quadratic region near $y = \hat{q}$, hypothesised to improve gradient flow. The 4.3% CRPS degradation suggests the kink serves a useful role in precisely identifying the quantile boundary.

The FILM backbone’s sensitivity to the loss landscape explains the larger degradations for Arctan and hybrid CRPS-Pinball variants: the per-block affine modulations amplify loss instabilities during early training.

The non-crossing cumsum architecture (217.06) forces strict quantile monotonicity by construction but reduces the model’s flexibility to fit individual quantile levels, slightly increasing CRPS. This confirms the finding of [5] that post-hoc sorting is preferable to hard architectural constraints for CRPS optimisation.

5.3 Evaluation and Discussion

5.3.1 Learning and Convergence Behaviour

Validation CRPS curves. All configurations exhibit a characteristic two-phase convergence: rapid improvement in epochs 1–5 (the model learns dominant daily and weekly seasonalities) followed by slower improvement in epochs 6–15 (tail quantiles and country-specific features are refined). Exogenous variants converge approximately 30% faster in absolute terms, as direct access to temperature eliminates the need to infer weather-driven demand from lagged patterns.

Gradient stability. The TCR and DBE variants exhibit higher gradient variance during the first 3 epochs, explaining their larger CRPS standard deviations across

seeds. The QCBC small-initialisation strategy (0.001 modulation scale) successfully prevents early instability while still allowing the quantile modulation to develop over training.

Learning rate schedule. The inverse square-root schedule with 400-batch warm-up proved critical for the exogenous models. Runs without warm-up (constant LR from epoch 1) diverged in 3 out of 5 seeds due to instability in the embedding lookup gradients during the first batches.

5.3.2 Consolidated Performance Ranking

Table 5.6 presents the consolidated ranking of all evaluated configurations by CRPS.

Table 5.6: Consolidated internal ranking by CRPS. Best in bold.

Rank	Configuration	CRPS	Cov-0.95	Δ CRPS
1	Stage 2-v2: Exog+Adaptive	172.21 $\pm$ 1.5	0.907	−18.5%
2	Stage 3-v2: Exog+Adaptive+Series	172.75 $\pm$ 1.8	0.895	−18.2%
3	Exog only (single seed)	168.46*	0.919	−20.3%*
4	Exog + Series (5 seeds)	174.09 $\pm$ 0.95	0.912	−17.6%
5	Series embedding only	206.51 $\pm$ 0.88	0.936	−2.2%
6	AQ-NBEATS paper baseline	211.22	—	—
7	DBE novel head	212.86 $\pm$ 1.50	0.924	+0.8%
8	QCBC novel head	213.22 $\pm$ 2.65	0.927	+0.9%
9	Non-crossing (cumsum)	217.06	0.922	+2.8%
10	Smooth Huber loss	220.34	0.898	+4.3%

*Single seed; multi-seed mean may differ.

5.3.3 Comparison with Published Benchmarks

Table 5.7 places our best results in the context of the full published benchmark from [16].

Table 5.7: Comparison with published benchmarks [16]. Our results in the lower section.

Model	CRPS	N-CRPS	MAPE (%)	vs. AQ-NBEATS
WaveNet	293.38	2.52	3.39	+38.9%
ES	325.80	2.64	3.48	+54.3%
AQ-ESRNN (paper)	195.94	1.72	2.32	−7.2%
AQ-NBEATS (paper)	211.22	1.84	2.47	—
Our results:				
Exog only (Exp 20, 1 seed)	168.46	—	2.11	−20.3%*
Stage 2-v2 (best multi-seed)	172.21	—	2.13	−18.5%
Stage 2-v2 + CQR (post-hoc)	172.21	—	—	(cov. 0.960)

*Single seed.

Our best multi-seed model (CRPS = 172.21) surpasses AQ-ESRNN (CRPS = 195.94) by approximately 12%. This is notable because AQ-ESRNN uses a hybrid architecture (exponential smoothing + dilated RNN) specifically designed for electricity demand, while our enhancement operates on the simpler N-BEATS backbone with added exogenous features. This result empirically confirms the hypothesis of Smyl et al. [16] that exogenous features are more impactful than the ESRNN architectural advantages on this benchmark.

5.3.4 Qualitative Analysis of Probabilistic Forecasts

Country-level observations. Exogenous features provide the most visible improvements for countries with high temperature sensitivity: Germany, France, and Norway. For these countries, winter cold spells drive demand spikes that purely autoregressive models predict too conservatively (wide intervals, poorly centred on the true value). The exogenous-augmented model narrows the PI while better centring it on the observed peak.

For countries with low temperature sensitivity (Iceland, Luxembourg), the exogenous improvement is smaller, consistent with the $v_y < 5\%$ variability coefficients for these countries.

Seasonal behaviour. Baseline models tend to under-cover in winter (when temperature effects are strongest and most variable) and over-cover in summer (more regular consumption patterns). The curriculum v2 model exhibits a more balanced seasonal coverage profile, with the beta-sampling Stage 1 improving winter tail coverage and the adaptive-sampling Stage 2 reducing summer over-coverage.

Quantile crossing. All models exhibit occasional quantile crossing before post-hoc sorting. Sorting reduces CRPS by approximately 2–5% for the AQ-NBEATS-based models in our study, consistent with the smaller benefit reported in [16] relative to AQ-ESRNN (where sorting improved CRPS by 30% due to the sigmoid nonlinearity).

5.3.5 Limitations of the Best Model

Coverage gap. Stage 2-v2 coverage of 0.907 falls below the 0.95 nominal target. This gap motivates the CQR post-processing step in Section 5.4.

Single-seed exogenous result. CRPS = 168.46 (Exp 20) is from a single training run. Given the ± 1.5 standard deviation observed in multi-seed exogenous experiments, the true multi-seed mean is likely in the range 166–171.

Reanalysis vs. operational weather. Our experiments use historical reanalysis data as a proxy for weather forecast inputs. In operational deployment, weather forecasts are available only with lead-time-dependent uncertainty. The true operational improvement may be smaller than our results suggest, particularly at the 36–48 hour horizon where temperature forecast errors are larger.

Holiday and event modelling. Public holidays produce systematic deviations from regular weekday patterns. The current model has no explicit holiday indicator, relying on the calendar embeddings to partially capture this effect. Residual errors on holidays represent the main remaining source of tail miscalibration.

5.4 Conformalized Quantile Regression Results

CQR post-processing is applied to the Stage 3-v2 model (Exog + Adaptive + Series embeddings), using a seasonal calibration/test split that maximises the challenge to distribution-shift robustness:

- **Calibration set:** January–June 2018 (winter and spring patterns).
- **Test set:** July–December 2018 (summer and autumn patterns).

This split ensures that the calibration data distribution is systematically different from the test distribution, constituting the hardest possible case for conformal calibration.

Table 5.8: CQR results. Calibration: Jan–Jun 2018. Test: Jul–Dec 2018. Target coverage: 0.95.

Method	CRPS	Coverage-0.95	Notes
Raw (no calibration)	129.49	0.914	Under-coverage (5 percentage points below target)
Symmetric conformal	129.31	0.914	Marginal gain from symmetric expansion
CQR	129.18	0.960	Meets and slightly exceeds 0.95 target

CQR achieves 96.0% empirical coverage despite the winter-to-summer calibration distribution shift, **confirming its robustness to seasonal distribution shift**. The CRPS reduction from 129.49 to 129.18 is minimal, indicating that the interval expansion required to achieve 95% coverage adds only a small amount of distributional width. This is a particularly positive result: naive conformal calibration often widens intervals excessively when the calibration and test distributions differ significantly.

The symmetric conformal method (which adds the same constant Δ to both bounds) achieves coverage of 0.914 — only marginal improvement over the uncalibrated model.

This demonstrates that the asymmetric CQR score (Eq. 1.11), which separately penalises upper and lower bound violations, is necessary for accurate calibration under the seasonal shift conditions tested here.

5.5 Provisional Chronogram

The project spanned 14 weeks from February to May 2026, structured in seven overlapping phases as shown in Table 5.9.

Table 5.9: Project Gantt chart — February to May 2026.

Phase	February				March				April				May
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1
Literature Review													
Environment Setup & Data Prep													
Baseline Implementation													
Advanced Architecture Dev.													
Web Application Development													
Testing & Validation													
Documentation & Reporting													

Conclusion and Perspectives

Summary of Work

This report presented a systematic experimental study of AQ-NBEATS enhancements for probabilistic short-term electricity demand forecasting on the EMHIRES MHLV benchmark, covering 35 European countries over a 48-hour forecast horizon. Starting from the published baseline CRPS of 211.22, we designed and evaluated more than twenty configurations across four complementary categories: exogenous feature and series-embedding augmentation, curriculum quantile training, novel architectural contributions, and post-hoc conformal calibration. The following conclusions summarise our principal findings.

1. Exogenous features are the single largest improvement. Integrating weather (temperature, humidity, pressure, wind speed) and calendar features reduces CRPS by approximately 20%, directly confirming the limitation identified in [16] and establishing exogenous integration as the highest-priority enhancement for any future work on this benchmark. Importantly, the improvement is robust: it persists across all multi-seed exogenous configurations evaluated (CRPS range 168–174) and generalises across both Curriculum v1 and v2 protocols. This single finding — that meteorological inputs matter more than architectural sophistication — is the most impactful practical contribution of this study.

2. Curriculum learning achieves the best multi-seed result. The two-stage Curriculum v2 protocol — beta sampling to establish tail calibration, followed by adaptive sampling to sharpen the distribution — achieves CRPS 172.21 (± 1.5) with coverage 0.907. This is the best reproducible result in the study and surpasses the AQ-ESRNN benchmark (195.94) by approximately 12%, despite the simpler N-BEATS architectural backbone. The curriculum ordering is critical: adaptive sampling applied without a calibrated initialisation consistently degrades both CRPS and coverage.

3. CQR reliably corrects miscalibration under distribution shift. Applied post-hoc to a trained quantile model, CQR achieves 96.0% empirical coverage under a challenging winter-to-summer calibration split, where the calibration and test distributions differ substantially. The marginal CRPS cost (129.49 to 129.18) demonstrates that the correction adds only minimal interval width, making CQR an efficient and practical post-processing step for any deployed quantile forecaster.

4. Novel architectural heads establish future baselines. QCBC, TCR, and

DBE all perform near the paper baseline (CRPS 212–217) without exogenous features. DBE achieves a best-seed CRPS of 210.99, marginally below baseline. While none of these contributions achieves a clear improvement in isolation, they provide principled frameworks and implementations for future research combining novel heads with exogenous inputs.

5. Interaction and ordering effects are critical. The most dramatic negative result in the study — monotonicity constraints on an exogenous model, CRPS = 239.22 — demonstrates that enhancement strategies can have strongly negative interactions when combined without curriculum ordering. This finding underscores the value of the ablation-driven methodology adopted in this study.

Future Perspectives

Several directions emerge naturally from the findings of this work:

Ensemble of best configurations. Combining Stage 2-v2, the single-seed exogenous model (Exp 20), and a series-embedding model via quantile-level median pooling could reduce variance and push CRPS below any individual model. The diversity of the three configurations (different training strategies, different feature sets) suggests low inter-model correlation and thus strong ensemble benefits.

Operational weather forecast inputs. The current study uses historical reanalysis data, which is a retrospective reconstruction of atmospheric conditions. Operational forecasting systems use *weather forecasts*, which carry their own uncertainty, particularly at the 24–48 hour lead times relevant to day-ahead electricity markets. Replacing reanalysis inputs with ensemble weather forecast products (e.g., ECMWF ENS or GFS) would more realistically simulate the operational deployment scenario and quantify the additional CRPS loss from weather forecast uncertainty.

Online CQR adaptation. The current CQR implementation uses a fixed calibration set and therefore cannot adapt to concept drift (e.g., a gradual climate shift or a structural change in demand behaviour). A sliding-window calibrator that continuously updates the conformity score quantile Δ using the most recent observations would maintain coverage guarantees throughout the year without requiring a fixed historical calibration period.

Transfer learning and per-country fine-tuning. The global cross-learning strategy trains a single model across all 35 countries simultaneously. An alternative is to pre-train globally (for data efficiency and generalisation) and then fine-tune per country on its own recent history. This approach could capture national demand idiosyncrasies — electricity market liberalisation, industrial restructuring, residential electrification — that the global model can only partially learn through series embeddings.

Holiday and extreme event modelling. The current system treats public holidays as regular days. Prior work has demonstrated that dedicated holiday models or rule-based corrections significantly reduce residual error on atypical days [17, 8, 2]. Integrating these corrections as a modular post-processing step, alongside explicit extreme

weather event flags (e.g., a temperature threshold indicator for heatwave detection), would address the main remaining source of tail miscalibration identified in this study.

Extension to other energy domains. The any-quantile framework and the curriculum learning methodology developed in this study are architecture-agnostic and loss-agnostic. Testing their transferability to related probabilistic forecasting tasks — solar photovoltaic generation, wind power output, natural gas demand — would demonstrate the generality of the contributions and provide additional benchmarks for the community.

Chapter A

Hyperparameter Tables and Supplementary Details

This annex provides complete hyperparameter tables for all baseline and enhanced model configurations, as well as supplementary technical details on the experimental setup.

A.1 AQ-NBEATS Baseline Hyperparameters

Table A.1: AQ-NBEATS-CAT baseline hyperparameter settings.

Hyperparameter	Value
Epochs	15
Loss function	Normalised MQN (Eq. 1.10)
Input normalisation	None
Hidden width (d_h)	1024
Residual blocks (R)	30
Layers per block (L)	3
Weight sharing	False
Lookback window (W , hours)	168
Forecast horizon (H , hours)	48
Batch size	1024
Optimiser	AdamW
Peak learning rate	0.0005
LR schedule	Inverse square-root, 400-batch warm-up
Quantile sampling	Uniform $U(0, 1)$
Dropout	0
Total parameters	74.1M

A.2 Exogenous Feature Model Hyperparameters

Table A.2: Additional hyperparameters for exogenous-augmented experiments.

Hyperparameter	Value
Continuous weather features	Temperature, Humidity, Pressure, Wind speed (4 features)
Calendar embedding dimensions	Hour: 4, Day-of-week: 4, Month: 4, Weekend: 2
Weather MLP hidden dim	64
Weather MLP output dim	168 (projected to lookback window size)
Feature scaling (continuous)	Standard scaling on training split
Calendar features	Learnable embedding tables (random normal init.)
Batch size (exog runs)	512 (memory constraint on RTX 3070 fp16)
Peak learning rate (exog)	0.0001 (reduced for stability)
Mixed precision (dev)	fp16 (RTX 3070); fp32 (A40 production)

A.3 Series Embedding Hyperparameters

Table A.3: Series embedding module settings.

Hyperparameter	Value
Number of countries	35
Embedding dimension (d_e)	32
Projection to input space	Linear layer, std 0.01 initialisation
Additive scale (α)	0.1 (baseline), 0.08 (combined with exog)

A.4 CQR Calibration Settings

Table A.4: CQR calibration settings.

Setting	Value
Base model	Stage 3-v2 (Exog + Adaptive + Series)
Calibration split	2018-01-01 to 2018-06-30
Test split	2018-07-01 to 2018-12-31
Target coverage ($1 - \alpha$)	0.95
Non-conformity score	CQR asymmetric (Eq. 1.11)
Conformity score quantile level	$(1 - \alpha)(1 + 1/N_{\text{cal}})$

A.5 Published Baseline Model Hyperparameters

For completeness, we reproduce the hyperparameter settings of all baseline models from Smyl et al. [16].

Table A.5: GluonTS neural baseline hyperparameters (from [16]).

Model	Hyperparameter	Value
All neural	Epochs	100
	Batch size	32
	Optimiser	Adam
	Learning rate	0.001
	Dropout	0.1
MLP	Hidden size	[40, 40]
	Activation	ReLU
DeepAR	Cell type	LSTM
	Hidden size	40
	Layers	2
	Scaling	True
Transformer	Model dim	32
	FF dim scale	4
	Attention heads	8
WaveNet	Residual channels	24
	Skip channels	32
	Stacks	1
	Epochs	200
	Learning rate	0.01
TFT	Hidden size	32
	Variable selection	True
	Attention heads	4

Chapter B		
Complete	Experiment	Log

Table [B.1](#) provides the complete experiment log across all 23 configurations evaluated in this study. Experiments are ordered by category and then by CRPS.

Table B.1: Complete experiment log. CRPS reported as mean $\pm$ std where multi-seed; single-seed values marked with *.

Exp	Category	Configuration	Seeds	CRPS	Cov-0.95
Paper	Baseline	AQ-NBEATS (published)	8	211.22	—
Exp 3	Baseline	Baseline reproduced	7	190.78 $\pm$ 1.42	0.942
Exp 19	Augment	Series embed only	7	206.51 $\pm$ 0.88	0.936
Exp 20	Augment	Exog only	1	168.46*	0.919
Exp 21	Augment	Exog + Series	5	174.09 $\pm$ 0.95	0.912
Exp 22	Augment	Exog+Series+Monotone	5	239.22	0.985
Exp 23	Curriculum	Exog+Series+Adapt(5q,0.8)	5	177.60	0.866
Exp 24	Curriculum	Exog+Series+Adapt(3q,1.0)	5	178.03	0.874
Exp 25	Curriculum	Exog+Series+Adapt(2q,1.2)	5	177.53	0.868
Exp 26	Curriculum	Curric v1 Stage 1	3	173.0	—
Exp 27	Curriculum	Curric v1 Stage 2	3	181.0	—
Exp 28	Curriculum	Curric v1 Stage 3	3	190.0	0.885
Exp 29	Curriculum	Curric v2 Stage 1 (Beta)	5	183.04 $\pm$ 1.9	0.947
Exp 30	Curriculum	Curric v2 Stage 2 (Adapt)	5	172.21$\pm$1.5	0.907
Exp 31	Curriculum	Curric v2 Stage 3 (+Embed)	5	172.75 $\pm$ 1.8	0.895
Exp 32	Curriculum	Curric v2 Stage 2 + CQR	1	129.18*	0.960
Exp 10	Novel arch	QCBC	5	213.22 $\pm$ 2.65	0.927
Exp 11	Novel arch	TCR	5	217.44 $\pm$ 5.42	0.923
Exp 12	Novel arch	DBE	5	212.86 $\pm$ 1.50	0.924
Exp 13	Novel arch	Non-crossing (cumsum)	1	217.06*	0.922
Exp 14	Alt loss	Smooth Huber pinball	1	220.34*	0.898
Exp 15	Alt loss	Smooth Arctan pinball	1	249.91*	0.916
Exp 16	Alt loss	Hybrid CRPS-Pinball	1	250.73*	0.913

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